Dynamic grid communications in the EU, with focus on Scandinavia

Research note · 5 August 2026 Desk research from public sources. Evidence-classed per the convention in `README.md`. Sources listed at the end.

The headline: this is no longer a proposal, it is a legal framework obligation

Directive (EU) 2024/1711 — the Electricity Market Design reform, in force 16 July 2024 — inserts **Article 6a into Directive (EU) 2019/944** requiring that:

> "The regulatory authority or another competent authority where a Member State has so provided **shall develop a framework** for transmission system operators and distribution system operators to offer the possibility of establishing **flexible connection agreements** in areas where there is limited or no network capacity availability."

The framework must ensure flexible connections do not, as a rule, delay reinforcement; that they convert to firm connections once the network is developed; and that they may be permanent where reinforcement would be inefficient. Agreements must specify **maximum firm and flexible injection and withdrawal capacity**, applicable network charges, duration, and expected date of full connection. System users connecting under one **must install certified power control systems**.

Transposition deadline: 17 January 2025 — already passed. The Commission opened infringement proceedings against 26 Member States in March 2025 for incomplete transposition of this directive.

The directive is marked "Text with EEA relevance."

What this changes for the Icelandic argument

The technical report currently argues that dynamic connections *should* exist, using California and Australia as precedent. The stronger argument is that Europe has already legislated them, that the instrument is EEA-relevant, and that the transposition deadline has passed on the EU side.

Three provisions map directly onto the Vík case:

- **"maximum firm and flexible withdrawal capacity"** — precisely the 120 kW-firm / non-firm-remainder split in SOW §11
- **"certified power control systems"** — the boundary controller and fail-safe requirement
- **"convert to firm once the network is developed"** — answers the reasonable DSO objection that non-firm access becomes a permanent dodge

Evidence class: Documented, from the consolidated directive text. **Not established by this research:** whether 2024/1711 has been incorporated into the EEA Agreement by Joint Committee decision, and what Iceland's implementation status is. Given the political history of the third energy package in Iceland, this must be checked directly — with Utanríkisráðuneytið, the EFTA Secretariat, or Raforkueftirlitið — before being asserted to RARIK. *Do not claim it is binding on Iceland today.*

Non-firm connections across Europe

Adoption is already majority practice. Non-firm/flexible connection agreements are implemented in **15 of the 29 states** cooperating through ACER. The EU Action Plan for Grids (2023) called for national frameworks; 2024/1711 made it an obligation.

Four models are distinguished in the literature:

Model	Mechanism
Capacity-limited	Reduced guaranteed capacity for faster/cheaper connection
Time-limited	Access restricted to defined periods
Dynamic operating envelopes	Real-time limits varying with network conditions
Fully flexible	Access entirely conditional

Reported effect: connection costs reduced by up to 80%, lead times halved. That figure is from the literature rather than an audited study, and should be cited as such — but even at a fraction of it, the economics dwarf the ISK 9.79 m/MW Landsnet capacity charge the report is built around.

Evidence class: Documented / Inference on the applicability of the 80% figure to Iceland.

Scandinavia, country by country

The Nordic picture

A 2025 Nordic Energy Research study identified **45 demand-side flexibility initiatives across the Nordics**, with ten studied in depth. Its headline findings:

- **Conditional connections are used or piloted in all Nordic countries**
- **Norway is the most active** — around 20 demand-side flexibility projects

- **Sweden is the only Nordic country with permanent DSF solutions**, the rest being pilots
- Time-of-use and power-based grid tariffs are being implemented region-wide

Norway — the furthest along on the connection question

Norway regulates conditional connection under the **NEM regulation (Forskrift om nettregulering og energimarkedet)**, §3-2 and related provisions. RME (the regulator, within NVE) proposed connection agreements with conditions as an explicit **alternative to network investment**. A grid company and a consumption customer may agree connection with terms for disconnection or limitation, allowing faster connection while avoiding unnecessary investment.

Amendments took effect **1 January 2025**, with further changes from **1 January 2026**. Connections ≥1 MW go through a maturity assessment and a capacity queue, with grid companies obliged to register request, offer, acceptance and capacity-availability dates.

Statnett publishes a standard framework agreement for connection on terms — *Rammeavtale om tilknytning på vilkår* — meaning this is contractual boilerplate in Norway, not a novel instrument.

Relevance to Vík: Norway has already solved the legal-instrument problem Iceland has not started. A named regulation, a standard contract, and a queue mechanism. This is the closest peer precedent available and far more persuasive to RARIK than California.

Sweden — permanent solutions and the flagship market

Sweden is the only Nordic country the study classes as having **permanent** demand-side flexibility solutions rather than pilots.

sthlmflex is the flagship: Svenska kraftnät (TSO) with **Ellevio** and **Vattenfall Eldistribution** (DSOs), **NODES** as independent market operator, addressing the Stockholm winter capacity gap. It is described as the only Swedish flexibility market testing **DSO–DSO** coordination as well as TSO–DSO, and it enabled exchange of transmission capacity subscriptions between connection nodes.

The Swedish regulator **Ei** commissioned a literature review on conditional grid connections (EiR2023:08) as part of its work on *villkorade avtal*.

Denmark — DSOs procuring flexibility to defer investment

Cerius and Radius together form Denmark's largest distribution network — **1.4 million customers, over a third of Danish consumption**. They are testing paying customers to relieve the grid, explicitly to defer infrastructure investment, and Danish DSOs are developing conditional connections in congested areas.

One number worth carrying into the Icelandic argument: in the Cerius-Radius network, constrained substations require demand reduction during only **approximately 3–4% of annual hours at most**. That is the quantified form of the report's §5.1 argument — capacity scarcity is a small number of hours, not a permanent condition, and pricing it as permanent is the error.

Finland

Helen Sähköverkko with Fingrid launched the first live local flexibility marketplace pilot in **winter 2025**.

Iceland — the gap

No Icelandic flexibility market, conditional connection framework, or DSO flexibility procurement was found in this research.

Landsnet presented on **Demand Side Response in Iceland** to the EU Clean Energy for EU Islands secretariat (2023). The document is a scanned/graphical PDF and could not be read by the tools available — **it should be retrieved and read manually; it is the single most relevant Icelandic source found.**

What is documented: Iceland's transmission system has extensive WAMPAC coverage with roughly **110 PMUs** and multiple wide-area control schemes. So the TSO has sophisticated monitoring and control at transmission level — the gap is at distribution, and specifically in the *market and contractual* layer, not the measurement layer.

Inference: Iceland is the only Nordic country without a conditional connection framework or flexibility market activity. That absence is itself the finding, and it is a stronger framing than "Iceland should try this" — it is "Iceland is the Nordic outlier, and the EU instrument that obliges the others is EEA-relevant."

The Netherlands — the operational benchmark

Not Scandinavian, but the most advanced regime in Europe and the closest match to the Vík expansion case.

Since **1 April 2025**, Dutch network operators offer a choice of connection and transport agreement types:

Instrument	What it does
Firm CTA (FA)	Full contracted capacity guaranteed at all times
Non-firm CTA-85 (NFA85)	Operator may reduce capacity within an **annual time budget**
Capaciteitsbeperkingscontract (CBC)	Capacity-limiting contract, separate from the 24/7 firm contract, typically running until local grid expansion completes
Time-based transmission right	Access by time of day
Time-block-based transmission right	Access within agreed time blocks — available on the **regional medium-voltage grid**

ACM approved the two new alternative transmission rights following a joint proposal from all Dutch grid operators via Netbeheer Nederland.

Why this matters for Vík: the NFA85 *annual time budget* is the mechanism that answers RARIK's likely objection —

that non-firm access is unbounded risk for the customer. A bounded, contractual, annual curtailment budget makes the product financeable. That is the design detail the Icelandic report currently lacks, and it is directly liftable.

Also relevant: **trials of time-dependent transport rights (TDTR)** offered by the Dutch TSO, and the observation that grid operators in some European countries already offer **CPOs specifically** flexible connection contracts allocating less capacity at morning/evening peak and more midday.

Protocols — what is actually deployed

This is where the research is least conclusive, and it matters for the north-side choice.

OpenADR in Europe. The Alliance's own material describes European activity as growing — UK, Germany, Netherlands, Croatia — and claims OpenADR is "fast becoming the most common standard for EV charging demand response applications." **Evidence class: vendor/consortium claim, not independently verified.** No evidence was found in this research of a *Nordic DSO* operating a production OpenADR VTN.

OSCP remains the protocol actually designed for the DSO→CPO case: a 24-hour forecast of available grid capacity, on 15-minute intervals, with the CPO adjusting charging profiles within it. The functional distinction found in the sources: **OSCP forecasts available capacity; OpenADR gives the DSO more direct control of equipment, including the ability to curtail directly.**

For a DSO issuing a binding envelope, that distinction favours OpenADR. For a DSO publishing a day-ahead capacity forecast, it favours OSCP. Worth stating explicitly in the report rather than treating OSCP's dormancy at OCA as disqualifying.

IEC 60870-5-104 vs IEC 61850. 104 remains in widespread use for substation→control-centre telemetry and "perfectly meets operators' current communication needs"; 61850 is the richer substation-automation standard and utilities expect RTU vendors to begin discontinuing 104 support. Neither is a DSO→CPO commercial interface. This supports the report's §8.4 boundary table: these live below the connection point, not across it.

OCPP 2.1 was released January 2025 with DER control and V2X support. That is directly relevant — it means the southbound leg of the architecture gains standardised DER control without waiting for anything, and it is worth checking whether Driivz or AMPECO have 2.1 on a roadmap.

Dynamic operating envelopes are described in the current literature as day-ahead signals from DSO to flexible customers such as EV aggregators, who adjust next-day scheduling. Note the cadence: **day-ahead, not real-time.** The Vík test is designed around near-real-time envelope changes, which is *more* demanding than mainstream European practice — worth knowing so the proof isn't undersold.

What this means for the Icelandic argument

Five changes I would make to the technical report on the strength of this:

1. **Lead with Article 6a, not California.** The report's §14.3 questions to Landsnet and §15 counterargument table both get materially stronger if the opening premise is "the EU has legislated this and the instrument is EEA-relevant" rather than "California does it."

1. **Replace the CSIP-AUS/Flex Connect precedents with Norway and the Netherlands as primary.** Norway for the legal instrument, the Netherlands for the product design. Keep California and Australia as secondary.

1. **Adopt the annual curtailment budget.** The Dutch NFA85 mechanism is the missing piece in §10's four-product split — it bounds the customer's downside and makes non-firm access financeable rather than merely cheaper.

1. **Use the Danish 3–4% figure.** "Constrained substations in Denmark's largest DSO network need demand reduction in 3–4% of annual hours" is the most compact statement of the report's central thesis found in this research.

1. **Reframe Iceland's position.** Not "Iceland should pilot this" but "Iceland is the Nordic outlier — conditional connections are used or piloted in every other Nordic country, Sweden has permanent solutions, Norway has a named regulation and a standard contract, and the EU instrument obliging the rest is EEA-relevant."

What I could not establish

Stated plainly so nothing here is over-claimed:

- **EEA incorporation status of Directive 2024/1711.** EEA-relevant ≠ in force in Iceland. Must be checked directly. **This is the single most important open question in this note.**
- **Landsnet's actual DSR position.** The 2023 presentation is a graphical PDF the tooling could not read. Retrieve and read manually.
- **Whether any Nordic DSO runs OpenADR in production.** Nothing found either way. The absence of evidence is not evidence of absence, but it does mean the "everyone uses OpenADR" claim cannot be made.
- **RARIK's or Orkustofnun's published position on flexibility**, if any.
- **Norwegian NEM §3-2 exact wording** — the provision was identified but the Lovdata text was not read in full.
- Three sources returned HTTP 403 or were unreadable scanned PDFs (Nordic Energy Research chapter pages, the Swedish Ei literature review, the Landsnet presentation). Their content here comes from search summaries rather than primary reading, and is marked accordingly.

Sources

EU law and policy

- Directive (EU) 2024/1711 — consolidated text, EUR-Lex
- Directive (EU) 2024/1711 — EUR-Lex record
- Commission infringement action on transposition, March 2025
- CEER Paper on Alternative Connection Agreements (2023)
- DSO Entity and ENTSO-E submit Joint Network Code on Demand Response
- ACER Recommendation 01/2025 on the Demand Response Network Code
- EPEX SPOT — More flex, less copper: the new Network Code on demand response
- EU DSO Entity Technical Vision, January 2025

Nordic

- Demand side flexibility in the Nordics — lessons from 10 case studies (Nordic Energy Research 2025:03)
- Current utilisation of flexibility in the Nordics (chapter)
- NVE/RME — connection agreements with conditions as an alternative to grid investment
- NEM regulation, Chapter 3 — supply and connection obligation (Lovdata)
- Statnett — Rammeavtale om tilknytning på vilkår
- DLA Piper — summary of NEM regulation changes
- Ei (Sweden) — Conditional grid connections: a literature review, EiR2023:08
- NODES — sthlmflex: market-based TSO-DSO coordination
- NODES — sthlmflex demonstrated a regional flexibility market
- JRC — Local Electricity Flexibility Markets in Europe
- Green Power Denmark — Cerius and Radius: shared operation, different challenges
- Danish DSOs buying flexibility from customers to defer investment

Netherlands

- ACM — more affordable transmission agreements for off-peak hours
- Van Doorne — alternative and flexible transmission capacity rights
- Enexis — Capaciteitsbeperkingscontract
- RAP — Gridlock in the Netherlands (2024)
- Contracting strategies for electrolyzers to secure grid connection: the Dutch case (arXiv)

Protocols

- OpenADR Alliance — focus on European flexible power
- Flexibility services based on OpenADR protocol for DSO level (Sensors, MDPI)
- Open Charge Alliance — Open Smart Charging Protocol (OSCP)
- AMPECO — the OCPP handbook
- E.DSO technical paper on V2G charging mechanisms, April 2025
- Comparison of IEC 60870-5-10x, DNP3 and TASE.2 with IEC 61850

Iceland

- Landsnet — Demand Side Response in Iceland (Clean Energy for EU Islands, 2023) — *graphical PDF, not machine-readable; retrieve manually*
- Government of Iceland — The State and Challenges of Energy Affairs (2022)

Academic and review

- Conditional connection agreements for EV charging: review, design and implementation (Renewable & Sustainable Energy Reviews)
- Increasing grid hosting capacity through non-firm connections and aggregator-mediated flexibility trades
- Evaluating non-firm grid connections for BESS: a co-optimization case study of the Netherlands
- Operating envelopes for the grid-constrained use of distributed flexibility in balancing markets (arXiv)